Dual Active Bridge Converter-I

Introduction

In the realm of power electronics, efficient and robust DC-DC converters are paramount, especially for high-power applications like electric vehicle charging infrastructure. Our previous discussions extensively covered the full-bridge isolated DC-DC converter, delving into its various schemes, including the Phase-Shifted Full-Bridge (PSFB) converter. While the PSFB offered advantages such as Zero Voltage Switching (ZVS) for reduced switching losses and improved EMI performance, it presented several limitations, particularly concerning bidirectional power flow and soft-switching range under varying load conditions.

This lecture introduces the Dual Active Bridge (DAB) converter, a sophisticated isolated DC-DC topology designed to overcome many of the PSFB's inherent drawbacks. The DAB converter is a highly versatile and efficient solution, particularly suited for high-power ranges, often exceeding 20-25 kilowatts. Its modular nature also allows for scalability, enabling the construction of very high-power chargers by connecting multiple DAB units in series or parallel. Understanding the DAB converter's structure, operation, and diverse modulation schemes is crucial for designing advanced power conversion systems capable of bidirectional power flow and enhanced performance.

Core Concepts

Term	Definition	Significance
`Dual Active Bridge (DAB)`	An isolated DC-DC converter topology featuring two active full-bridge circuits, one on the primary side and one on the secondary side of a high-frequency transformer.	Enables bidirectional power flow and offers superior control flexibility compared to passive rectifier-based converters.
`Phase-Shifted Full-Bridge (PSFB)`	A full-bridge converter where power transfer is controlled by phase-shifting the switching signals of diagonal switches, primarily on the primary side.	A precursor to DAB, it introduced ZVS but lacked inherent bidirectional power flow and had limited soft-switching range.

Term	Definition	Significance
`Zero Voltage Switching (ZVS)`	A soft-switching technique where a switch turns on when the voltage across it is zero, significantly reducing turn-on switching losses.	Crucial for high-frequency operation, improving efficiency and thermal management in power converters.
`Flux Walking`	A phenomenon in transformers where a DC offset in the primary current causes the magnetic flux to drift, potentially leading to core saturation.	Can cause transformer saturation and failure; requires control strategies like peak current mode control to mitigate.
`Generic Switch`	A conceptual switch representing a power semiconductor device (e.g., MOSFET with body diode or IGBT with freewheeling diode) capable of bidirectional current flow.	Simplifies circuit analysis by abstracting the specific device implementation, focusing on its switching behavior.
`Leakage Inductance (L)`	The inductance inherent in a transformer due to imperfect magnetic coupling between windings, or an additional series inductor.	Acts as the primary energy transfer element in DAB converters, facilitating power flow control through phase shifts.
`Single Phase Shift (SPS) Modulation`	A DAB control strategy where power flow is regulated by varying only the phase shift between the primary and secondary full-bridge voltages.	The simplest modulation scheme, offering fundamental control over power direction and magnitude.
`Phase Angle Ratio (delta*)`	A normalized representation of the phase shift angle (delta) between the primary and secondary bridge voltages, typically scaled by 180 degrees.	Simplifies calculations for time delays corresponding to phase shifts in switching waveforms.

Detailed Analysis

1. Limitations of the Phase-Shifted Full-Bridge (PSFB) Converter

Before diving into the DAB converter, it's essential to understand the shortcomings of its predecessor, the PSFB, which the DAB aims to address. The PSFB, while offering ZVS, presented several challenges:
* **No Bidirectional Power Flow:** Due to the diode-based rectifier on the secondary side, power could only flow in one direction (from input to output). This is a significant limitation for applications requiring energy storage or regenerative braking.
* **Limited Range of Soft Switching:** Achieving ZVS, especially for the lagging leg switches, was challenging under light load conditions. The leakage inductance required sufficient energy to discharge the output capacitors, which was not always available.
* **Large Inductive Filter on the Secondary Side:** The PSFB typically required a large inductive filter on the secondary side, contributing to the converter's size, weight, and impacting its power density.
* **Flux Walking:** This phenomenon, caused by DC offset in the transformer primary current, could lead to core saturation. While solvable with peak current mode control, it added complexity to the control scheme. Non-idealities in devices (e.g., different RDS_on) or gate driver delays could contribute to this issue.

Introduction

The DAB converter emerges as a robust solution to these challenges. As its name suggests, it features two "active" bridges:
* **Structure:**
* A full-bridge inverter on the primary side, connected to the input voltage (Vin). This bridge consists of four generic switches (S1, S2, S3, S4).
* A high-frequency transformer, providing isolation and voltage scaling (NP:NS).
* An inductor (L) in series with the transformer, which can be the transformer's leakage inductance or an additional external inductor. This inductor is crucial for energy transfer.
* A full-bridge inverter/rectifier on the secondary side, connected to the output (Vout) and consisting of four generic switches (S5, S6, S7, S8).
* An output capacitor (C) and load (RL) on the secondary side.
* **Generic Switches:** The switches are represented generically, implying they can be implemented using MOSFETs with body diodes or IGBTs with freewheeling diodes, allowing for bidirectional current flow.
* **Active Bridges:** Both the primary and secondary bridges are "active" because they can actively apply voltages (between 0 and Vin, or 0 and Vout) in a switching

average sense. This contrasts with passive diode rectifiers, enabling bidirectional power flow.
* **Inductor Role:** The inductor 'L' is key. By applying different voltages across it from the primary and secondary bridges, current can be forced to flow in either direction, facilitating power transfer. The value of 'L' influences the current slope and power transfer capability.

3. Control Variables and Modulation Schemes in DAB Converters

The DAB converter offers significant control flexibility through multiple phase-shift variables:
* **Control Variables:**
* **`Delta (δ)`:** The phase shift between the primary full-bridge (FB1: S1-S4) and the secondary full-bridge (FB2: S5-S8). This is the primary control for power flow direction and magnitude.
* **`Theta_1 (θ1)`:** The phase shift between the two half-bridges within the primary full-bridge (e.g., between S1/S2 and S3/S4). This effectively controls the duty ratio of the voltage applied from the primary bridge (VA1B1).
* **`Theta_2 (θ2)`:** The phase shift between the two half-bridges within the secondary full-bridge (e.g., between S5/S6 and S7/S8). This controls the duty ratio of the voltage applied from the secondary bridge (VA2B2).
* **Modulation Schemes:** By manipulating these variables, various modulation strategies are possible:
* **Single Phase Shift (SPS) Modulation:** Only `delta` is varied. This is the simplest scheme, controlling power flow by shifting the phase between the two full bridges.
* **Dual Phase Shift (DPS) Modulation:** Involves varying `delta` along with either `theta_1` or `theta_2`, or even `theta_1` and `theta_2` simultaneously. This allows for more refined control over the magnitude of the applied voltages.
* **Triple Phase Shift (TPS) Modulation:** All three variables (`delta`, `theta_1`, and `theta_2`) are controlled. This offers the highest degree of control, potentially leading to reduced RMS currents and conduction losses.
* **Variable Frequency (VF) Modulation:** Any of the above schemes can be combined with variable switching frequency (fsw). Changing fsw alters the impedance (ωL) of the inductor, providing another degree of control over current flow. TPS with Variable Frequency (TPS-VF) is particularly noted for achieving very low RMS current values.

4. Single Phase Shift (SPS) Modulation Waveform Analysis

To understand DAB operation, let's analyze the waveforms for SPS modulation, assuming `theta_1` and `theta_2` are zero (i.e., switches in each half-bridge operate

with 50% duty cycle, and diagonal switches turn on/off simultaneously).
* **Switching Sequence Assumption:**
* Primary side: S1 and S4 turn on/off together; S2 and S3 turn on/off together.
* Secondary side: S5 and S8 turn on/off together; S6 and S7 turn on/off together.
* **Voltage Waveforms:**
* **`VA1B1` (Primary Bridge Output):**
* When S1 and S4 are ON, VA1B1 = +Vin.
* When S2 and S3 are ON, VA1B1 = -Vin.
* This results in a square wave alternating between +Vin and -Vin, with a period of Ts.
* **`VA2B2` (Secondary Bridge Output, referred to primary):**
* A phase shift `delta` is introduced between the primary and secondary bridge switching.
* If S6 and S7 are ON, VA2B2 = -Vout/N (referred to primary).
* If S5 and S8 are ON, VA2B2 = +Vout/N (referred to primary).
* This also results in a square wave, but phase-shifted by `delta` relative to VA1B1.
* **Phase Angle Ratio (`delta*`):**
* A phase shift of `delta` degrees corresponds to a time delay of `(delta / 180°) * (Ts / 2)`.
* We define `delta* = delta / 180°` for convenience, so the time delay is `delta* * (Ts / 2)`.
* **Inductor Current (`IL`) Waveform:**
* The current `IL` flowing through the inductor is determined by the voltage difference across it: `V_L = VA1B1 - (VA2B2 / N)`.
* The slope of `IL` is `V_L / L`.
* **During the initial phase (e.g., 0 to `delta* * Ts/2`):** If VA1B1 is +Vin and VA2B2 (referred) is -Vout/N, the voltage across L is `Vin + Vout/N`. The current `IL` rises with a steep positive slope.
* **During the next phase (e.g., `delta* * Ts/2` to `Ts/2`):** If VA1B1 is +Vin and VA2B2 (referred) is +Vout/N, the voltage across L is `Vin - Vout/N`. The current `IL` rises with a less steep positive slope (assuming Vin > Vout/N).
* The waveform then becomes symmetrical in the negative half-cycle.
* The exact shape of the current waveform depends on the relative magnitudes of Vin and Vout/N.
* **Avoiding Flux Walking:** To prevent DC offset in the transformer primary current and subsequent flux walking, the current waveform must be symmetrical. This means:
* The peak current in the positive half-cycle (I1) should be equal in magnitude but opposite in sign to the peak current in the negative half-cycle (I3).
* The current at the beginning of a half-cycle (I0) should be equal in magnitude but opposite in sign to the current at the end of that half-cycle (I2).

* Specifically, `I2 = -I0` and `I1 = -I3`, ensuring `I4 = I0`. This guarantees no net DC component in the transformer current.

Key Takeaways

- [ ] Dual Active Bridge (DAB) converters enable bidirectional power flow, unlike traditional PSFB converters with diode rectifiers.
- [ ] DAB converters utilize two active full-bridge circuits, one on the primary and one on the secondary side of a transformer.
- [ ] The series inductor (L) is critical for energy transfer and power flow control in DAB converters.
- [ ] DAB converters offer multiple control variables: `delta` (phase shift between bridges), `theta_1` (duty ratio of primary bridge), and `theta_2` (duty ratio of secondary bridge).
- [ ] Various modulation schemes exist, including Single Phase Shift (SPS), Dual Phase Shift (DPS), and Triple Phase Shift (TPS), which can also be combined with variable frequency operation.
- [ ] SPS modulation controls power by varying only the phase shift (`delta`) between the primary and secondary bridge voltages.
- [ ] Symmetrical current waveforms are essential in DAB converters to prevent flux walking and transformer saturation.
- [ ] DAB converters are well-suited for high-power applications (e.g., 20-25 kW) and can be modularly scaled.

Conclusion

The Dual Active Bridge converter represents a significant advancement in isolated DC-DC conversion, directly addressing the limitations of earlier topologies like the Phase-Shifted Full-Bridge. Its inherent capability for bidirectional power flow makes it indispensable for modern applications such as electric vehicle charging, grid-tied energy storage, and renewable energy systems, where power needs to flow both to and from the load. The flexibility offered by its multiple control variables and diverse modulation schemes allows for optimized performance, including reduced switching losses through soft-switching and minimized conduction losses by shaping current waveforms.

As we move forward, understanding the detailed design considerations for the DAB, including the selection of the inductor 'L' and the precise derivation of current waveforms under various operating conditions, will be crucial. The ability to control power flow, manage efficiency, and ensure transformer health by preventing flux walking positions the DAB converter as a cornerstone technology for the next

generation of high-power, high-efficiency, and highly reliable power conversion systems.